Securing understanding of addition and subtraction with fractions

Task A: Addition and subtraction involving mixed numbers

1) By calculating the value of each side complete the inequality statement by placing a < or > between them.

a) $4\frac{3}{4} + 2\frac{1}{5} \;>\; 3\frac{1}{2} + 3\frac{2}{5}$

b) $4\frac{4}{5} - 2\frac{3}{4} \;<\; 3\frac{1}{2} - 1\frac{2}{5}$

c) $1\frac{5}{8} + 2\frac{3}{4} \;>\; 2\frac{5}{6} + 1\frac{1}{3}$

2) Find the missing value in each diagram. Give your answer as a mixed number in its simplest form.

a) $3\frac{3}{8}$ $\boxed{4\frac{31}{40}}$ $1\frac{2}{5}$

b) $\boxed{\frac{3}{4}}$ $2\frac{5}{12}$ $1\frac{2}{3}$

c) $2\frac{5}{12}$ $\boxed{4\frac{1}{12}}$ $1\frac{2}{3}$

d) $\boxed{1\frac{39}{40}}$ $3\frac{3}{8}$ $1\frac{2}{5}$

e) $1\frac{5}{6}$ $\boxed{5\frac{7}{12}}$ $3\frac{3}{4}$

f) $\boxed{\left(-1\frac{11}{12}\right)}$ $1\frac{5}{6}$ $3\frac{3}{4}$

3) Find the missing digits:

a) $1\frac{2}{7} + 2\frac{\boxed{2}}{3} = 3\frac{20}{21}$

b) $2\frac{2}{9} - 1\frac{\boxed{1}}{6} = 1\frac{1}{18}$

c) $1\frac{\boxed{5}}{6} + 2\frac{3}{4} = 4\frac{7}{12}$

Task B: Efficient calculation strategies

1) Which of these are equal to $9\frac{3}{4}$?

a) $3\frac{1}{4} + 3\frac{1}{4} + 3\frac{1}{4}$ ✓

d) $\frac{51}{3} - \frac{29}{4}$ ✓

g) $\frac{90}{5} - 7\frac{3}{4}$ $\boxed{-\frac{1}{2}}$

b) $5\frac{5}{20} + 3\frac{4}{5}$ $\boxed{+\frac{7}{10}}$

e) $12\frac{1}{5} - 2\frac{9}{20}$ ✓

h) $7\frac{1}{6} - 3\frac{1}{4} + 5\frac{2}{3}$ $\boxed{+\frac{1}{6}}$

c) $10\frac{1}{6} - \frac{1}{2}$ $\boxed{+\frac{1}{12}}$

f) $2\frac{1}{4} + 5\frac{1}{2} + 1\frac{3}{4}$ $\boxed{+\frac{1}{4}}$

i) For those that do not have an answer of $9\frac{3}{4}$, what do you need to add or subtract so that they do?